Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

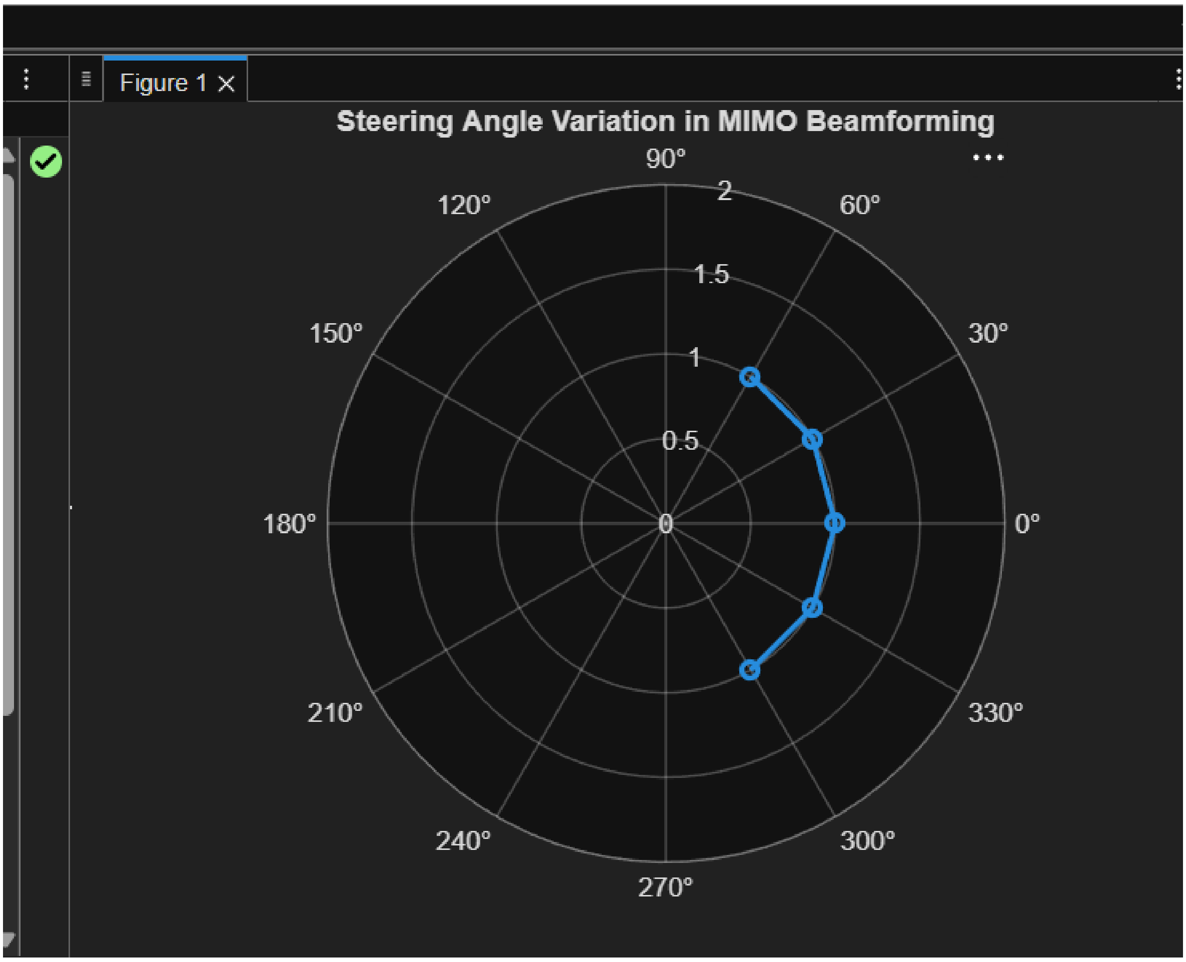
OBJECTIVE 1:

CODE:

```
clc;
clear;
close all;

angle = [-60 -30 0 30 60];   % Steering Angles (degrees)
gain = ones(size(angle));    % Normalized Gain

figure
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)
title('Steering Angle Variation in MIMO Beamforming')
```



OBJECTIVE2:

CODE:

```
clc;
clear;
close all;

angle = [-60 -30 0 30 60];   % Steering Angles (degrees)
gain = [9 12 15 12 9];       % Array Gain (dB)

figure
```

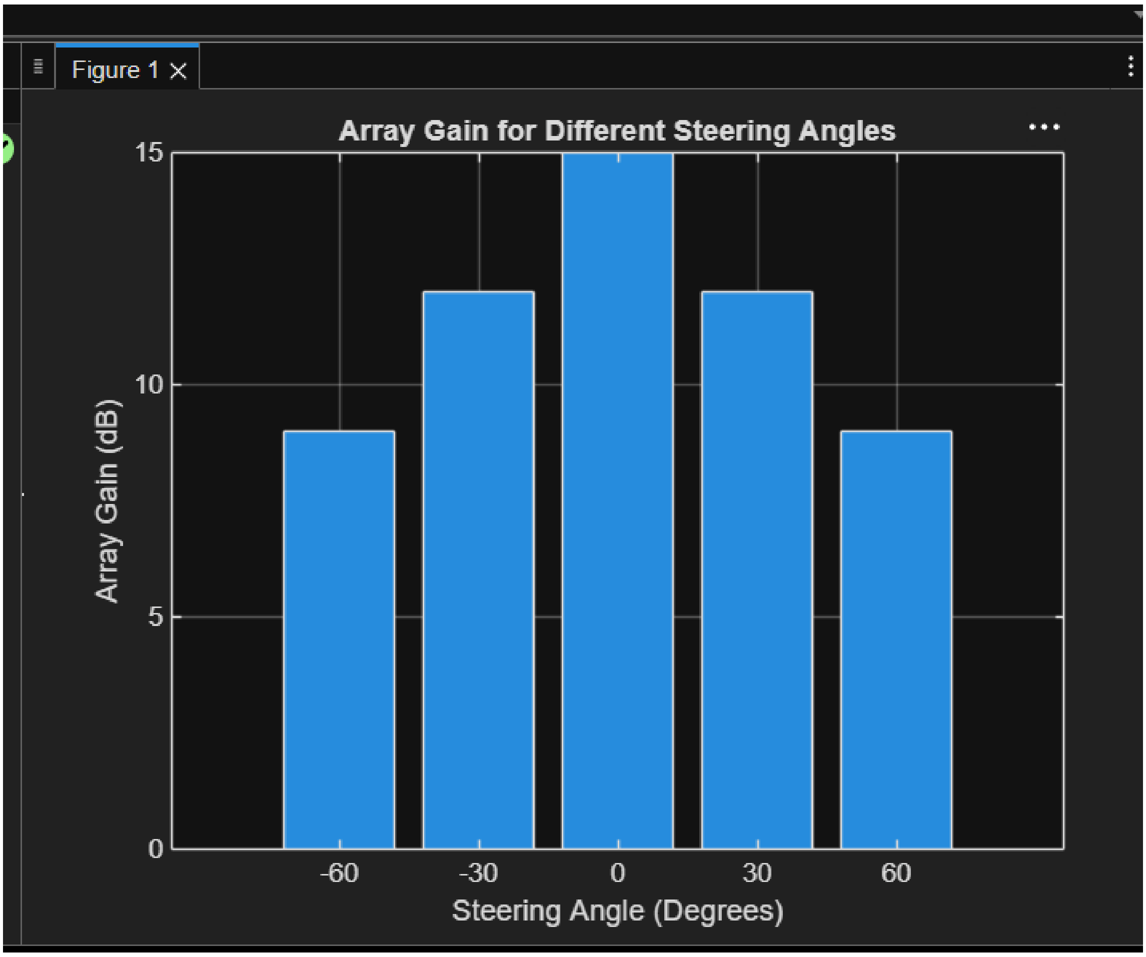
bar(angle, gain)

grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')



OBJECTIVE 3:

CODE:

clc;

clear;

close all;

```
desired = [-60 -30 0 30 60];      % Desired Steering Angles (degrees)
actual  = [-58 -32 1 29 62];      % Actual Beam Directions (degrees)

error = abs(desired - actual);    % Beam Direction Error

figure
plot(desired, error, 'o-', 'LineWidth', 2)
grid on

xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')
```

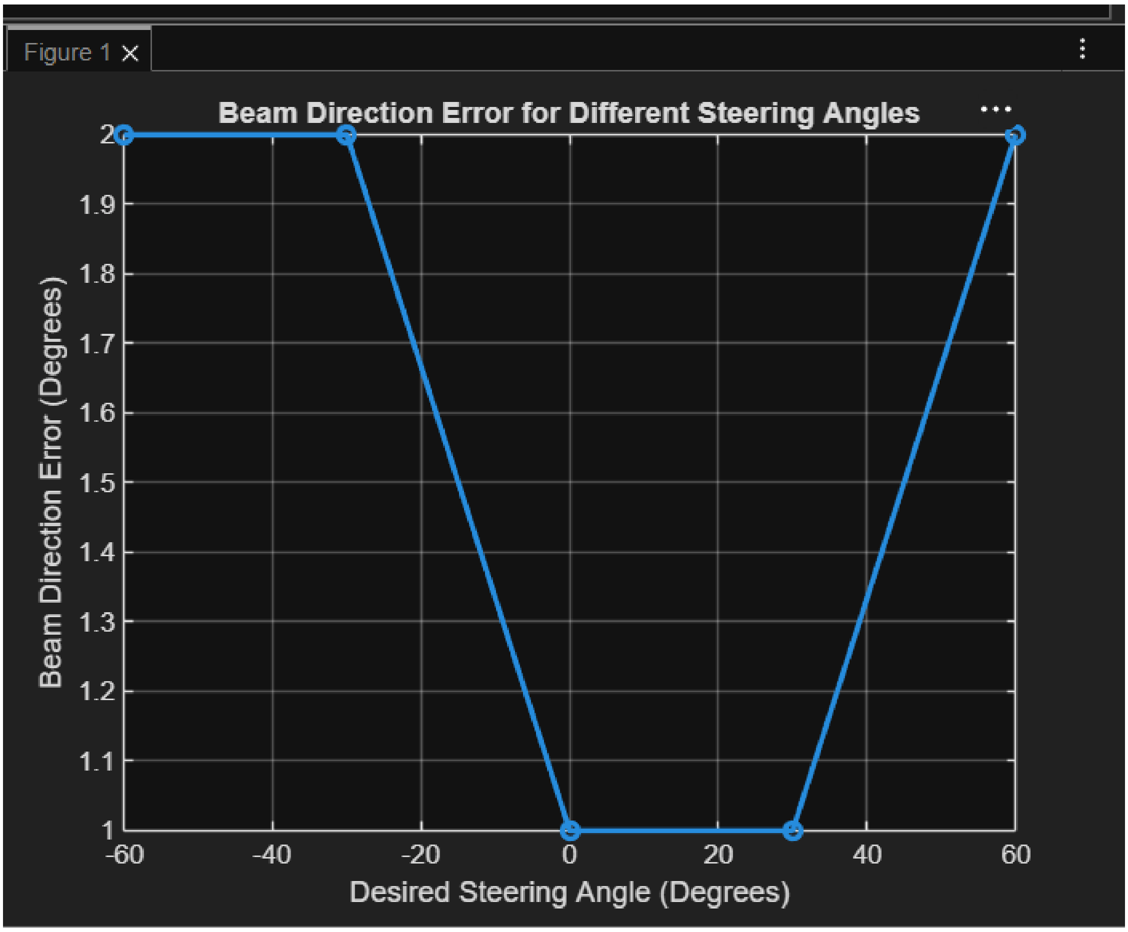


Figure 1
Beam Direction Error for Different Steering Angles
Beam Direction Error (Degrees)
2
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
-60
-40
-20
0
20
40
60
Desired Steering Angle (Degrees)